VW2 Subgoal Distillation Report

Push-T hard-stop evaluation on branch `vw2_subgoal_distill`

April 13, 2026

Executive Summary

This round implemented the continuous-subgoal branch, trained the TeacherOracle policy, and ran the required 50-rollout Push-T evaluation with 100-step episodes and execute-actions-per-plan in {1, 2}. The branch failed Gate A immediately. TeacherOracle reached **0.0% success** on both execute settings, so the required hard stop was applied. StudentFrozen and StudentJoint were not trained or evaluated.

The failure pattern is concrete: TeacherOracle improved offline action MSE slightly relative to BC (0.02137 vs. 0.02281), but world success remained zero and mean episode reward became worse than BC on both execute settings. This is sufficient evidence to stop the future-conditioned Push-T branch at this stage.

Scope Completed

- Created branch `vw2_subgoal_distill`.
- Added a history encoder that consumes the last 4 observations, proprio, and previous actions.
- Added a future encoder with a 32-dimensional continuous bottleneck and no tokenization.
- Added a horizon-conditioned subgoal predictor.
- Reused the current action decoder as `Actor(context, subgoal)` inside a new subgoal system.
- Added entrypoints for teacher training, student training, joint training, and evaluation.
- Added offline diagnostics for variance, covariance, retrieval, shuffled retrieval, and action gap.
- Added world evaluation with 50 rollouts, 100 steps, execute sweep {1, 2}, per-episode CSV export, and rollout video export.
- Added dataset support for real Push-T HDF5 loading through the stable-worldmodel backend.
- Verified code health with `compileall` and an 8-test unit suite in the project virtual environment.

Key Files Added or Updated

Path	Purpose
`vw2_directact/models/history_encoder.py`	Encodes 4-step observation and action history into a context vector.
`vw2_directact/models/future_bottleneck.py`	Encodes future observations into a 32-dimensional continuous subgoal.
`vw2_directact/models/subgoal_predictor.py`	Predicts the continuous subgoal from context and horizon.

`vw2_directact/data/subgoal_dataset.py`	Provides Push-T history/future batches and real HDF5 loading support.
`vw2_directact/subgoal_system.py`	Defines teacher, student, and joint subgoal training logic.
`vw2_directact/train/train_teacher_oracle.py`	TeacherOracle training entrypoint.
`vw2_directact/train/train_student_predictor.py`	StudentFrozen training entrypoint.
`vw2_directact/train/train_joint_subgoal.py`	StudentJoint training entrypoint.
`vw2_directact/train/eval_subgoal_policy.py`	Offline and world evaluation with gate checks and artifact export.
`vw2_directact/utils/rollout.py`	Online subgoal policy for BC, oracle, and student rollout modes.
`vw2_directact/tests/test_subgoal.py`	Unit coverage for dataset, model shapes, and training smoke tests.

Experiment Configuration

- Dataset: Push-T from `C:\Users\yiche\.stable-wm\pusht_expert_train.h5`
- Runtime device: local RTX 4070 8GB
- Remote 4090/5090 spaces: not used for this run
- Teacher initialization checkpoint: `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_falsification_oracle\action\last.ckpt`
- Teacher output directory: `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_subgoal_distill_round1\teacher_oracle`
- Evaluation setting: 50 rollouts, 100 max steps, execute-actions-per-plan in {1, 2}
- Subgoal setup: history length = 4, subgoal dimension = 32, plan horizon = 8
- Training precision: `bf16-mixed`

Teacher training completed through epoch 2 (0-indexed), with the final logged validation metrics at step 599:

Metric	**Value**
`val/loss`	0.0113455
`val/teacher_action_mse`	0.0226943
`val/subgoal_batch_variance`	2.8563×10^{-5}

Results

Offline and World Metrics

Model	Offline MSE	Exec-1 Success	Exec-2 Success	Notes
BC	0.022812	0.0%	0.0%	latency = 39.69 ms
TeacherOracle	0.021369	0.0%	0.0%	offline improves, world stays dead

Model	Exec-1 Mean Reward	Exec-2 Mean Reward
BC	-23644.90	-24199.08
TeacherOracle	-48110.09	-36832.53

Gate Evaluation

Gate	Status	Rule / Outcome
Gate A	**FAIL**	TeacherOracle success had to be at least 90% on execute-1 and execute-2. Measured result: 0.0% on both.
Gate B	**NOT RUN**	Not run because Gate A failed and the branch hard-stopped.
Gate C	**NOT RUN**	Not run because Gate A failed and the branch hard-stopped.
Gate D	**NOT RUN**	Not run because Gate A failed and the branch hard-stopped.

Interpretation

The teacher policy learned a slightly better offline action fit than BC, but that improvement did not translate into control. World success stayed at zero, and reward regressed sharply. The most defensible conclusion from the finished run is:

> The continuous-subgoal teacher did not produce a usable Push-T policy, so the future-conditioned branch should stop here on Push-T.

This conclusion follows the stated protocol exactly. No extra model capacity was added. No extra ablations were run after the gate failure. StudentFrozen and StudentJoint were intentionally skipped.

Artifacts Produced

- BC rollout videos: `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_subgoal_distill_round1\eval_50rollouts_100steps\videos_execute_1`
- Subgoal evaluation summary: `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_subgoal_distill_round1\eval_subgoal_50rollouts_100steps\summary.json`

- Per-episode CSVs: `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_subgoal_distill_round1\eval_subgoal_50rollouts_100steps\BC` `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_subgoal_distill_round1\eval_subgoal_50rollouts_100steps\TeacherOracle`
- TeacherOracle rollout videos: `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_subgoal_distill_round1\eval_subgoal_50rollouts_100steps\TeacherOracle\videos_execute_1`
- Teacher checkpoints and logs: `C:\Users\yiche\PycharmProjects\lab_code_26spring\research_code\vw2_directact_outputs\pusht_subgoal_distill_round1\teacher_oracle`

Final Decision

Stop the whole future-conditioned Push-T branch on this subgoal-distillation variant. The completed experiment already falsifies the branch under the required gate sequence.